Transfer: thermal energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose an energy-transfer model and communicate its limits.

SUCCESS CRITERIA - I can...

1. Distinguish warming, phase change and work.
2. Use correct signs and units.
3. Explain why a model result depends on its assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A heat engine absorbs Q_{hot}=2400 J and rejects Q_{cold}=1680 J each cycle. Find work and efficiency.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q = mc\,\Delta T$; $Q = mL$; $\Delta U = Q - W_{by}$; $\eta = W_{out}/Q_{hot}$. Select the applicable model and supplied constants. State heat-loss, phase-change and cycle assumptions.

Worked reasoning: A system takes 120 J of heat, does 50 J of work and increases internal energy by 70 J. An energy model must include transfers and storage rather than treating all heat as useful work.

Your turn: Draw its system boundary, input, output and storage change.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

A hot block and cool water are isolated together. Draw initial heat flow and describe equilibrium without inventing a final temperature.



Hot block



Cool water

Each outline represents the labeled sample or container.

Marzano check - circle: ① ② ③

04 Apply the physics

A beaker contains 0.10 kg of ice at 0°C. Melt it and warm the water to 10°C. Use L_f = 334,000 J/kg and c_{water} = 4200 J/(kg·°C). Show both stages in the energy account and find the total energy required.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

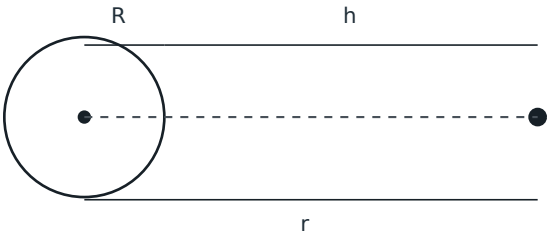
05 Explain your reasoning

An engine absorbs 500 J and produces 150 J work in a cycle. Find rejected heat and explain why an energy balance alone does not establish that any temperature pair can achieve that efficiency.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simulated source has GM=288 m³/s². At r=4 m, find escape speed and state two assumptions.
Label the distances or energy levels used in your reasoning.



Marzano check - circle: 1 2 3

07 Exit check

A heat engine between 500 K and 300 K absorbs 1000 J. Find ideal maximum work and explain why actual work usually falls below it. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.5 kg sample with c=400 J/(kg·K) warms 2 K while doing 60 J of work on surroundings. Find heat input under ΔU=mc ΔT and state one condition required for that model. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.